Normalization of neural responses

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1 Methods

1.1 Notation

Let:

- n: the number of sample stimuli.
  - n_f: the number of texture families. Here, assume noisy texture families are omitted, so $n_f = 35$
  - n_s: the number of samples within each texture family.
  - n_r: the number of repetitions of each specific sample.
  - n_b: the number of blank stimuli.
- k_i: the number of recording sites for region i.
  - If needed, this can also be sectioned by array/simultaneously recorded sites.
- t: the number of 10 ms bins of multi-unit activation (MUA) counts.

The raw MUAs for a given region are contained in a tensor of size $n \times k_i \times t$. For most analyses, we will summarize over t to produce a two-dimensional matrix $\mathbf{X}$. Then, $X_{i,j}$ is a specific element, $\mathbf{X}_{-,i}$ is a $n \times 1$ column vector (the responses of one site to all stimuli), and $\mathbf{X}_{i,-}$ is a $1 \times k_i$ row vector (the responses elicited across all sites by one stimulus).

1.2 Mahalanobis distance

Let P be some probability distributions on $\mathbb{R}^k$ with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. The Mahalanobis distance of points $\mathbf{x}, \mathbf{y}$ with respect to P is

$$d_M(\mathbf{x}, \mathbf{y}, P) = \sqrt{(\mathbf{x}-\mathbf{y})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}-\mathbf{y})}.$$

If $\boldsymbol{\Sigma}$ is positive semi-definite, the covariance can be decomposed as $\boldsymbol{\Sigma} = \mathbf{S}^\top \mathbf{S}$, where $\mathbf{S}$ is the standard deviation matrix. So, the Mahalanobis distance is equivalently

$$d_M(\mathbf{x}, \mathbf{y}, P) = \|\mathbf{S}^{-1}(\mathbf{x}-\mathbf{y})\|,$$

or the Euclidean distance after a whitening transformation.

1.2.1 Usage in Hotelling's two-sample *t*-test

Let $P \sim \mathcal{N}_k(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, i.e., a k-dimensional multivariate Normal distribution. Then, let $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{n_x}$ and $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_{n_y}$ be independent samples from P. Let each sample be a column vector. We can then define the sample mean vectors

$$\overline{\mathbf{x}} = \frac{1}{n_x}\sum_{i=1}^{n_x}\mathbf{x}_i, \quad \overline{\mathbf{y}} = \frac{1}{n_y}\sum_{i=1}^{n_y}\mathbf{y}_i,$$

and the sample covariance matrices as

$$\hat{\boldsymbol{\Sigma}}_{\mathbf{x}} = \frac{1}{n_x - 1}\sum_{i=1}^{n_x}(\mathbf{x}_i - \overline{\mathbf{x}})(\mathbf{x}_i - \overline{\mathbf{x}})^\top, \quad \hat{\boldsymbol{\Sigma}}_{\mathbf{y}} = \frac{1}{n_y - 1}\sum_{i=1}^{n_y}(\mathbf{y}_i - \overline{\mathbf{y}})(\mathbf{y}_i - \overline{\mathbf{y}})^\top.$$

The unbiased pooled covariance matrix estimate is the weighted average

$$\hat{\boldsymbol{\Sigma}} = \frac{(n_x - 1)\hat{\boldsymbol{\Sigma}}_{\mathbf{x}} + (n_y - 1)\hat{\boldsymbol{\Sigma}}_{\mathbf{y}}}{n_x + n_y - 2},$$

which is simply $\left(\hat{\boldsymbol{\Sigma}}_{\mathbf{x}} + \hat{\boldsymbol{\Sigma}}_{\mathbf{y}}\right)/2$ when $n_x = n_y$.

Though not used in our analyses, the resulting Hotelling's two-sample t-squared statistic has the distribution

$$t^2 = \frac{n_x n_y}{n_x + n_y}(\overline{\mathbf{x}} - \overline{\mathbf{y}})^\top \hat{\boldsymbol{\Sigma}}^{-1}(\overline{\mathbf{x}} - \overline{\mathbf{y}}) \sim T^2(k, n_x + n_y - 2).$$

1.2.2 Effect of normalization on Mahalanobis distance calculations

As mentioned previously, the Mahalanobis distance is equivalent to the Euclidean distance after whitening with $\boldsymbol{\Sigma}^{-1/2}$. Thus, any data processing that can be "undone" by whitening will not change the outcome. For instance, centering and scaling with respect to the sites will result in the same value.

Let $\mathbf{b} \in \mathbb{R}^k$ be some arbitrary constant vector and $\mathbf{m} \in \mathbb{R}^k$ be some non-zero constant vector. If we take

$$\begin{aligned}\mathbf{x}' &= \begin{bmatrix} m_1 x_1 + b_1 & m_2 x_2 + b_2 & \cdots & m_k x_k + b_k \end{bmatrix}^\top, \\ \mathbf{y}' &= \begin{bmatrix} m_1 y_1 + b_1 & m_2 y_2 + b_2 & \cdots & m_k y_k + b_k \end{bmatrix}^\top,\end{aligned}$$

then transform the assumed covariance matrix of the new distribution P accordingly with

$$\tilde{\boldsymbol{\Sigma}} = \operatorname{diag}(\mathbf{m})\boldsymbol{\Sigma}\operatorname{diag}(\mathbf{m}),$$

then the term in the Mahalanobis distance simplifies to

$$\begin{aligned}(\mathbf{x}' - \mathbf{y}')^\top \tilde{\boldsymbol{\Sigma}}^{-1}(\mathbf{x}' - \mathbf{y}') &= (\mathbf{x} - \mathbf{y})^\top \operatorname{diag}(\mathbf{m}) \left(\operatorname{diag}(\mathbf{m})\boldsymbol{\Sigma}\operatorname{diag}(\mathbf{m})\right)^{-1} \operatorname{diag}(\mathbf{m})(\mathbf{x} - \mathbf{y}) \\ &= (\mathbf{x} - \mathbf{y})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \mathbf{y}).\end{aligned}$$

Though the sample setting estimates the covariance rather than relying on the known P, this invariance to centering and scaling still holds. That is,

$$
\begin{aligned}
\hat{\mathbf{\Sigma}}_{\mathbf{x}'} &= \frac{1}{n_x - 1} \sum_{i=1}^{n_x} (\mathbf{x}'_i - \overline{\mathbf{x}}')(\mathbf{x}'_i - \overline{\mathbf{x}}')^\top \\
&= \frac{1}{n_x - 1} \sum_{i=1}^{n_x} \operatorname{diag}(\mathbf{m})(\mathbf{x}_i - \overline{\mathbf{x}})(\mathbf{x}_i - \overline{\mathbf{x}})^\top \operatorname{diag}(\mathbf{m}) \\
\mathbf{x}' - \mathbf{y}' &= \operatorname{diag}(\mathbf{m})(\mathbf{x} - \mathbf{y})
\end{aligned}
$$

The effect of centering and scaling on $\hat{\Sigma}$ can still be reduced to multiplication by a diagonal matrix.

1.2.3 Pairwise between-family distances

When calculating the distance between each pair of texture families, each sample image and each repetition can be considered a draw from the underlying distribution.